PENETRATION TESTING REPORT

FOOTPRINTING & RECONNAISSANCE PHASE

W2-PM1 | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B084J - Networkwalks
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Modules completed	W2-PM1 (Footprinting with Multiple Kali Tools)
Client/Target	Networkwalks (networkwalks.com), the target specified in the course exercise
Permission secured from client?	Yes
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2-5:** Upcoming

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1). Reconnaissance is the first phase of any security test: before touching a target, a tester collects as much public information as possible about it. The tools used here only read information that the target has already made public, so they do not attack the target.

All commands were run in Kali Linux inside a VirtualBox virtual machine built in Week 1. Every step below includes the exact command used, the result I observed, and a short note on why the finding matters from an attacker's point of view. Screenshots are provided as evidence in Section 8. This is the Week 2 part of my ongoing internship program at Networkwalks.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used for all reconnaissance activities.
whois	Find domain registration details (registrar, dates, name servers).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP).
nslookup	Resolve the domain name to its IP address using DNS.

Tool	Purpose
curl	Read the HTTP response headers of the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate DNS records (SOA, NS, MX, A, TXT/SPF, SRV).

4. Activities Performed

I performed reconnaissance against the `networkwalks.com` domain using six Kali Linux tools: **whois, whatweb, nslookup, curl, wafw00f and dnsrecon**. Each tool was used to collect a different type of information about the target.

4.1 Domain Registration (whois)

whois queries the public domain registration record to find who owns the domain, when it was registered and which name servers it uses.

```
whois networkwalks.com
```

Observations:

- Registrar: GoDaddy.com, LLC (IANA ID 146).
- Created 2019-11-06, last updated 2025-11-12, registry expiry 2027-11-06.
- Name servers: NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM.
- Domain status shows delete, renew, transfer and update prohibited (registrar lock enabled).
- DNSSEC: unsigned.

Attacker's view: The name servers point to HostGator, so the hosting provider is learned immediately. Dates and abuse contacts can help with planning and social engineering.

4.2 Web Technology Fingerprinting (whatweb)

whatweb identifies the web server, CMS, plugins, frameworks and IP address of the website.

```
whatweb networkwalks.com
```

Observations:

- Web server: Apache.
- CMS: WordPress 7.1.4 with the WordPress Download Manager plugin 3.3.76.
- Other technologies: jQuery 3.7.1, HTML5 and Google Tag Manager.
- Server IP: 192.232.216.135 (United States).
- Contact email exposed: info@networkwalks.com.
- The HTTP site redirects to HTTPS (301 Moved Permanently).

Attacker's view: Exact software and versions can be looked up in vulnerability databases to find known issues. The output also leaks the server IP and an email address.

4.3 DNS Resolution (nslookup)

nslookup resolves the domain name to its IP address using DNS.

```
nslookup networkwalks.com
```

Observations:

- networkwalks.com resolves to 192.232.216.135.
- The answer is non-authoritative and was returned by the local DNS server 192.168.31.1.

Attacker's view: Knowing the real IP allows direct scanning of the server, searching for other sites on the same IP, and mapping the infrastructure.

4.4 HTTP Response Headers (curl)

curl was used to read the HTTP response headers, which show the server banner, status, cookies and redirects. The screenshot shows the command run as curl -i, which prints the headers followed by the page content.

```
curl -i https://networkwalks.com
```

Observations:

- Status: HTTP/2 200, server: Apache.
- Cache header x-nginx-cache: WordPress shows part of the caching stack.
- The link header exposes the WordPress REST API endpoint /wp-json/.
- Cookie __wpdm_client is set with the secure and HttpOnly flags.
- referrer-policy and permissions-policy headers are present; strict-transport-security, content-security-policy and x-frame-options do not appear in the captured response.

Attacker's view: Headers fingerprint the stack and reveal entry points such as /wp-json/ without loading the full page.

4.5 Web Application Firewall Detection (wafw00f)

wafw00f (v2.4.2) detects whether a Web Application Firewall protects the target site.

```
wafw00f networkwalks.com
```

Observations:

- The site https://networkwalks.com is behind a ModSecurity (SpiderLabs) WAF.
- Detection took 2 requests.

Attacker's view: Knowing a WAF is present shapes the attack: naive attempts will be blocked or logged, so an attacker must adapt or try to bypass it.

4.6 DNS Enumeration (dnsrecon)

dnsrecon enumerates DNS records. My first attempt without the -d option failed with an "unrecognized arguments" error; the command below is the corrected version.

```
dnsrecon -d networkwalks.com
```

Observations:

- SOA and NS records: ns6135.hostgator.com (50.87.144.87) and ns6136.hostgator.com (192.232.216.131).
- Both name servers expose the DNS software version Bind 9.16.23-RH.
- MX record: mail.networkwalks.com (192.232.216.135). A record: 192.232.216.135.
- SPF record: v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all, plus a Google site-verification TXT record.
- 8 SRV records for _autodiscover._tcp pointing to cpanelemaildiscovery.cpanel.net on port 443.
- No answer to the DNSSEC query, which matches the unsigned result from whois.

Attacker's view: The DNS footprint reveals the mail server, DNS software version, SPF policy and cPanel hosting setup. Each record helps build a picture of the email and hosting infrastructure.

5. Risk Analysis / Impact

Based on the information collected during the footprinting activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology and version information exposed	WhatWeb identified WordPress 7.1.4 and WordPress Download Manager 3.3.76	Attackers may use exposed technology and version information to identify software requiring further security review	● **Medium**
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	● **Low**
3	HTTP technical information exposed	Curl returned response headers that exposed the Apache server banner, caching headers and the /wp-json/ REST API endpoint	May assist technology fingerprinting and further enumeration	● **Low**
4	WAF technology identifiable	Wafw00f identified ModSecurity (SpiderLabs)	Reveals information about the web application's security architecture	● **Low**
5	DNS infrastructure information exposed	DNSRecon identified name servers, mail server, SPF/TXT records, cPanel SRV records and DNS software version Bind 9.16.23-RH	DNS information can help build a broader infrastructure and email-setup profile	● **Medium**
6	DNSSEC not enabled	WHOIS showed DNSSEC: unsigned and DNSRecon received no answer to its DNSSEC query	DNS responses are not cryptographically signed, which removes one layer of DNS integrity protection	● **Low**
7	Contact email address exposed	WhatWeb identified info@networkwalks.com on the website	Exposed email addresses can be used for phishing or social engineering	● **Low**

Risk level key: ● Critical ● Medium ● Low

The risks above are observations from the footprinting exercise, not confirmed vulnerabilities.

The practical exercise involved only information gathering. No exploitation or vulnerability validation was performed as part of this module.

Therefore, the presence of information such as a software version, IP address or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Reduce exposed technology information**
   Remove unnecessary version details, such as the WordPress generator meta tag, so that the site reveals less about its CMS and plugins.
2. **Keep software updated**
   Keep WordPress, WordPress Download Manager and all other plugins up to date and review them against current security advisories.
3. **Review HTTP headers**
   Review response headers to remove unnecessary technical information, and consider adding standard hardening headers such as Strict-Transport-Security, Content-Security-Policy and X-Frame-Options, which were not present in the captured response.

4. **Restrict the WordPress REST API where not needed**
   Review what the /wp-json/ endpoint exposes and restrict anything that is not required publicly.
5. **Review DNS records and DNS software exposure**
   Hide the DNS software version (Bind 9.16.23-RH) where possible, remove unused records, and review the SPF policy, which currently ends in ~all (soft fail).
6. **Consider enabling DNSSEC**
   If supported by the registrar and DNS host, enable DNSSEC to protect the integrity of DNS responses.
7. **Properly configure and monitor the WAF**
   Keep the ModSecurity WAF enabled and tuned, since it already blocks naive attacks.
8. **Limit publicly exposed contact information**
   Use a contact form instead of a plain email address on the website to reduce phishing and spam exposure.
9. **Perform security testing with authorization**
   Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship, I completed a practical footprinting and reconnaissance activity against the networkwalks.com domain.

I used six Kali Linux tools to collect information about the target. I learned how whois can provide domain and name-server information, whatweb can identify web technologies, nslookup can resolve domain names, curl can inspect HTTP headers, wafw00f can identify a WAF, and dnsrecon can reveal DNS, mail and service records. Together these tools built a full profile of the target using only public information.

The exercise showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment, and defenders can run the same tools on themselves to see what an attacker would see.

I also learned that tools need the correct syntax: my first dnsrecon attempt failed until I used the -d option, and reading the usage message helped me fix it. In addition, I learned that technical findings should be documented clearly, explaining what was performed, what was discovered, what it means and how the risk can be reduced.

Finally, I learned that reconnaissance must always be performed within an authorized scope. These activities were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

Figure 1: whois networkwalks.com

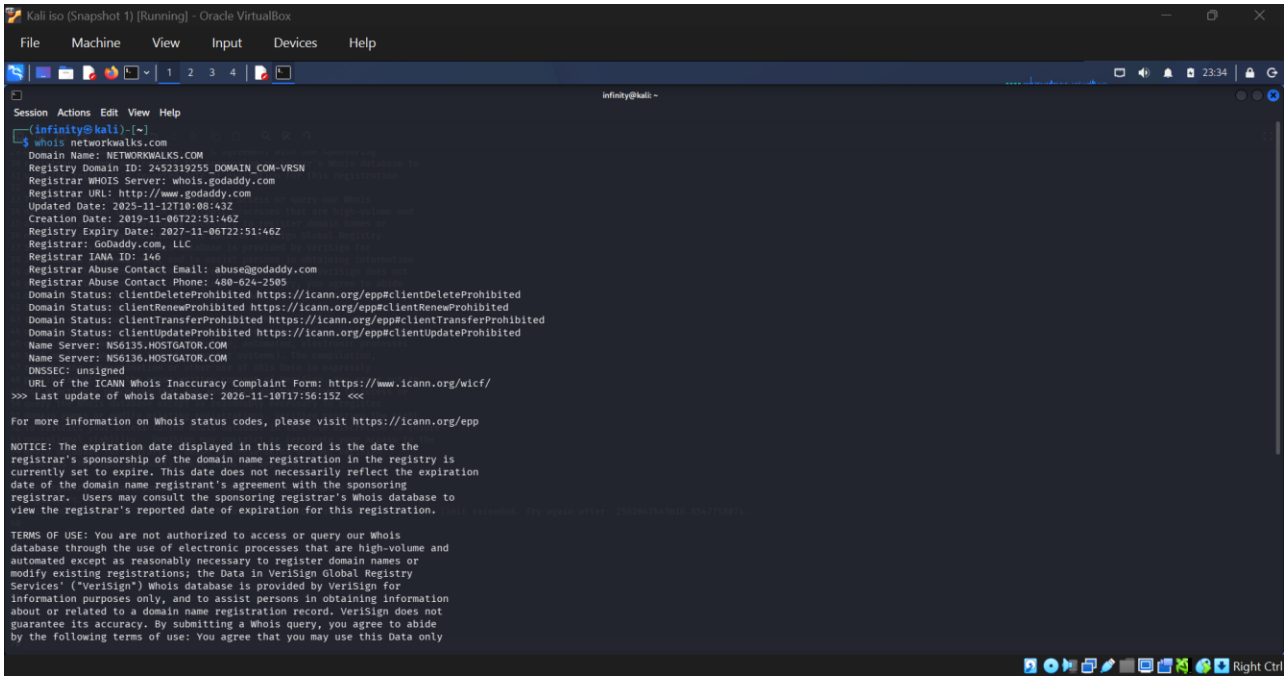


Figure 2: whatweb networkwalks.com



Figure 3: nslookup networkwalks.com (shown below the whatweb output)



Figure 4: curl -i https://networkwalks.com



Figure 5: wafw00f networkwalks.com



Figure 6: dnsrecon -d networkwalks.com

